UPS and downs

Anti-virus protection is dandy, but how do you safeguard against losing valuable data to a black out? As an undercover agent I am generally prepared for anything. But the other day I was caught unawares. Hard at work on a particular assignment, my PC suddenly went on the blink thanks to an electrical failure. As I sat there in the darkness hoping I hadn't lost any data, I resolved to upgrade my kit with the best protection for such occurrences—a UPS.

Chancing on a computer shop in Andheri, I asked for a UPS for my PC. He asked me where I lived. Taken aback, I asked him why he wanted to know. His answer: "Most suburbs in north Mumbai face acute power problems and hence a UPS is recommended. Otherwise an ordinary power strip would be more than sufficient." "But how will a power strip help when a brown-out occurs?" I argued. His logic was that brown-outs don't occur too often. Well, he was mistaken and I soon found that very few PC and electronics dealers knew much about the vagaries of power. A brown-out happens when lower than normal line voltage is supplied by the power company. A power line voltage reduction of 8 to 12 per cent is usually considered a brown-out.

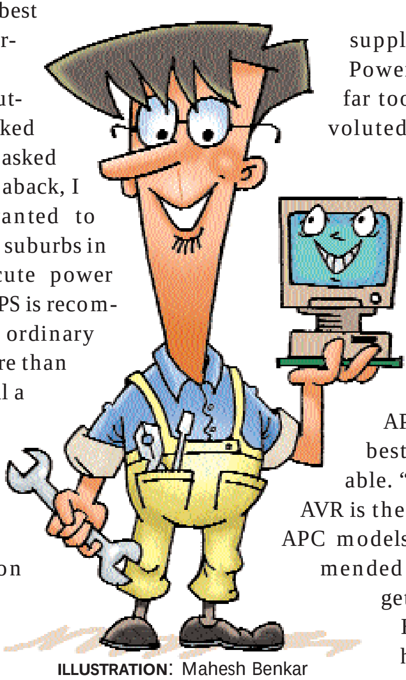

ILLUSTRATION: Mahesh Benkar

Hoping to find a better solution at an electrical hardware shop, I walked into one and asked about how best to guard my PC in case of a brown-out. This chap knew what he was talking about. He told me how brown-outs cause unregulated power supply to the computer. Power supply problems are far too complicated and convoluted errors due to erratic power supply can creep into almost every single PC operation.

My next stop was Lamington Road. The first dealer I spoke to recommended a UPS from APC. He said they are the best and the most affordable. "The APC Back-UPS 500 AVR is the most popular of all the APC models and is widely recommended and sold too. You can get it for about Rs 4,500". He added that unless I had a 17-inch or higher monitor with printers, scanners and more than two ATAPI devices (more than one CD-ROM and one hard drive), this UPS would suffice. Otherwise I'd have to go in for one with a higher VA rating.

At my next stop I told the vendor that I needed a UPS but didn't know how to calculate the VA rating. He confessed that he didn't know either, but he did know that 500-650 VA is enough for must PC users. True. But if you have a 19-inch monitor and several other power consuming hardware then 1 KVA would be the minimum you'd require.

Another vendor told me that the TVSE line of UPSs was doing well too. I asked him how he'd rate the local (home made, assembled, or unbranded) UPSs. According to him, local manufacturers were pretty good when it came to day-to-day working but they lacked consistency and had flaky support mechanism.

One vendor told me that for home usage, an offline UPS is the only way to go. An online UPS would be too expensive and is recommended only if you are running an expensive PC for critical applications and data. He also rightly added that an offline UPS has a switchover time (from mains to battery power) of milliseconds, which most power supplies can handle very easily without resetting the PC.

Another vendor said the ideal environment to run any electronic equipment was to run a voltage stabiliser along with a reliable offline UPS, or use a line interactive online UPS because most UPS systems operate your computer directly off your 'dirty' power, except during periods of lost power or brown-outs. Unfortunately, most UPS systems have limited spike suppression and virtually no interference filtering capabilities. Also, during brown-outs your UPS may continuously alternate between the power from the mains and the UPS power and this discharges batteries and thrashes the PC at the same time. But he also said that the APC UPS was definitely a cut above the rest and highly recommended. He suggested that one should keep aside 10 per cent of the PC budget towards voltage stabilisers, UPS, power strips etc. Lastly, he advised on ensuring that all power connections are properly earthed and not loose.

SHOPPING TIP

- To buy a UPS all you need to do is add up the total power draw of your equipment.

First, decide which pieces of equipment need UPS support. Typically, only the CPU and monitor are supported to cut down on power draw to the UPS.

- If the power draw is expressed in Amps then multiply that value by your nominal line voltage, 230V. If the power draw is expressed in Watts, multiply by a factor of 1.3 to 1.4 for VA load.
- Example: 250 Watt power supply (x 1.4) = 350 VA load (assuming that your monitor is drawing power from the power supply of the PC). Add a measure of safety and 500 VA is enough for most PCs.

Monitor Hunting

Imagine buying a 17-inch monitor for less than Rs 10,000. Yes, It's very much possible now! A majority of the popular brands and makes have reduced their prices and have finally broken the Rs 10,000 barrier. With a little bargaining you might even get a 15-inch monitor for less than Rs 6,500.

A little hunting will also fetch you amazing prices for some of the brands. The lowest we found for a 17-inch was Rs 8,000. So go sniffing!